UNIT-I


HUMAN VISION & COMPUTER VISION
SEEING THE WORLD, INTELLIGENTLY

Topics :

- Introduction.
- Difference b/w Human Vision and Computer Vision
- Comparison of Biological Vision and Computer Vision
- Real World Example
- Processing Components
  - Image Acquisition
  - Image Processing
  - Image Analysis

Computer Vision in Deep Learning

- **Computer Vision (CV)** is a **field of Artificial Intelligence (AI)** that enables computers to **see, understand, analyze, and interpret images and videos**, **just like humans use their eyes and brain to understand the world.**
- Unlike **humans, computers do not naturally understand images**. They see **images as numbers (pixel values).** Deep Learning helps **computers learn patterns from these pixels and recognize objects automatically.**
- **Computer Vision** is the ability of a **computer to acquire, process, and understand visual information from images or videos** using **Deep Learning algorithms.**

- With the introduction of **Deep Learning**, especially **Convolutional Neural Networks (CNNs),** **computers can automatically learn useful features from raw images without manual feature engineering**.
- Computer Vision was mainly based with **image processing algorithms and methods.**
- The **main process** of **computer vision** was **extracting the features of the image.** ie, **detecting the color, edges, corners, and objects** was the **first step** to do when performing a **computer vision task.**
- This has significantly improved the performance of **image recognition, object detection, medical diagnosis, autonomous driving**, **Motion Tracking, Coloring Black and White Images** and many other applications.

HUMAN VISION



Real-World Object
Objects in the real world



Eye
Captures light from the objects



Retina
Captures light and converts it into electrical signals



Optic Nerve
Transmits electrical signals to the brain



Brain (Visual Cortex)
Processes and interprets visual information



Recognition
Identifies objects and understands them



Decision
Makes decisions and actions

COMPUTER VISION



Real-World Object
Objects in the real world



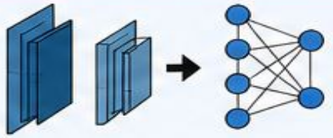
Camera
Captures image



Digital Image
Converts image into digital data (pixels)



Preprocessing
Resizing, filtering, normalization, etc.



CNN Feature Extraction
Automatically learns features from data



Classification / Detection / Segmentation
Identifies or locates objects



Prediction & Decision
Outputs result and takes action

Comparison of Biological Vision and Computer Vision

Biological Vision	Computer Vision
Uses the **human eye and brain** to **understand the world.**	Uses **cameras and AI algorithms** to **understand images and videos.**
Input is **light reflected from objects.**	Input is **digital images or video captured by a camera**.
The **eye (retina) captures the image**.	The **camera/image sensor captures the image**.
The **optic nerve sends signals** to the brain.	The **image is transferred as digital data (pixels)** to the computer.
The **brain (visual cortex) processes** and interprets the image.	A **Deep Learning model (CNN**) processes and interprets the image.
Learns naturally from **experience throughout life**.	Learns from **large labeled datasets during training**.
Can **recognize objects even with incomplete information.**	Performance **depends on the quality of data and training.**
Very energy efficient (**human brain uses about 20 W).**	Requires significant computational power **(CPU/GPU).**
Makes **decisions using reasoning, memory, and experience.**	Makes decisions based on **learned patterns and probabilities.**

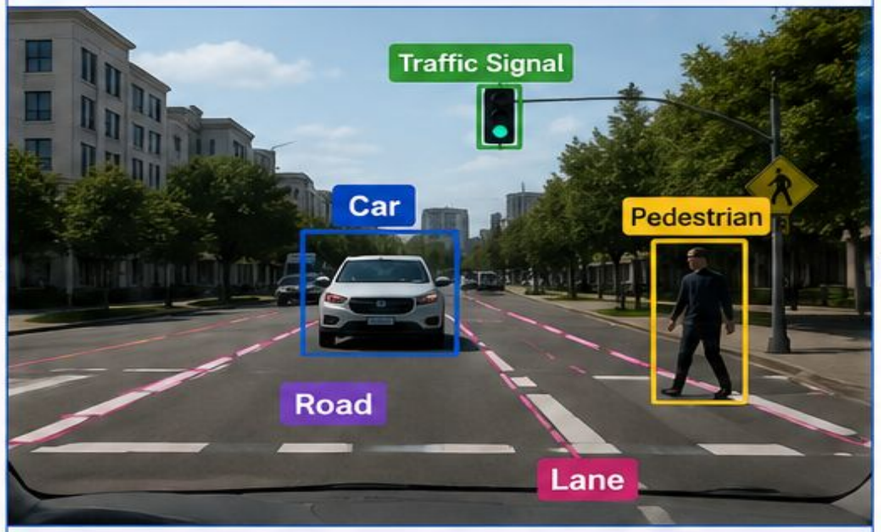
REAL-WORLD EXAMPLE : SELF-DRIVING CAR

1. CAMERA CAPTURES THE ROAD



The camera mounted on the car captures continuous images of the road and surroundings.

2. DEEP LEARNING MODEL IDENTIFIES OBJECTS AND ELEMENTS



The Deep Learning model (CNN) analyses the image and identifies:

- Road
- Car
- Pedestrian
- Traffic Signal
- Lane

3. VEHICLE DECIDES THE ACTION

Based on the identified information, the self-driving car decides the best action.

- **STOP** (If obstacle or red signal)
- **MOVE** (If path is clear)
- **TURN LEFT** (If required by navigation or road conditions)

4. EXECUTION

The vehicle follows the decision by controlling:

- Steering
- Brakes
- Acceleration

HOW IT WORKS (FLOW)

Camera captures real-time images → Deep Learning model processes the image → Objects and elements are identified → Best action is decided → Vehicle executes the action

In short: Camera → Deep Learning Model → Object Detection → Decision Making → Action

EXAMPLE: BIOLOGICAL VISION vs COMPUTER VISION

BIOLOGICAL VISION

A person sees a traffic signal.

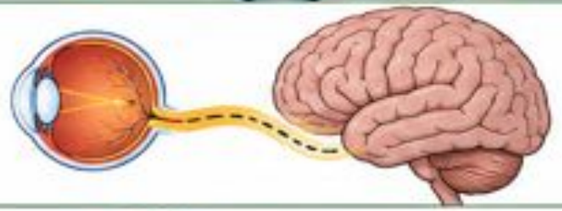
1. Eyes capture the light.



2. Retina converts light into electrical signals.



3. Optic nerve sends signals to the brain.



4. Brain recognizes Red Signal.



5. Person decides to stop.



OUTPUT: PERSON STOPS

COMPUTER VISION

A self-driving car sees a traffic signal.

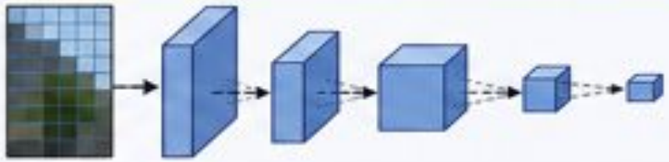
1. Camera captures the image.



2. Image is converted into digital pixels.



3. CNN extracts features.



4. AI identifies Red Traffic Signal.



Prediction:
Red Traffic Signal

5. Vehicle decides to stop.

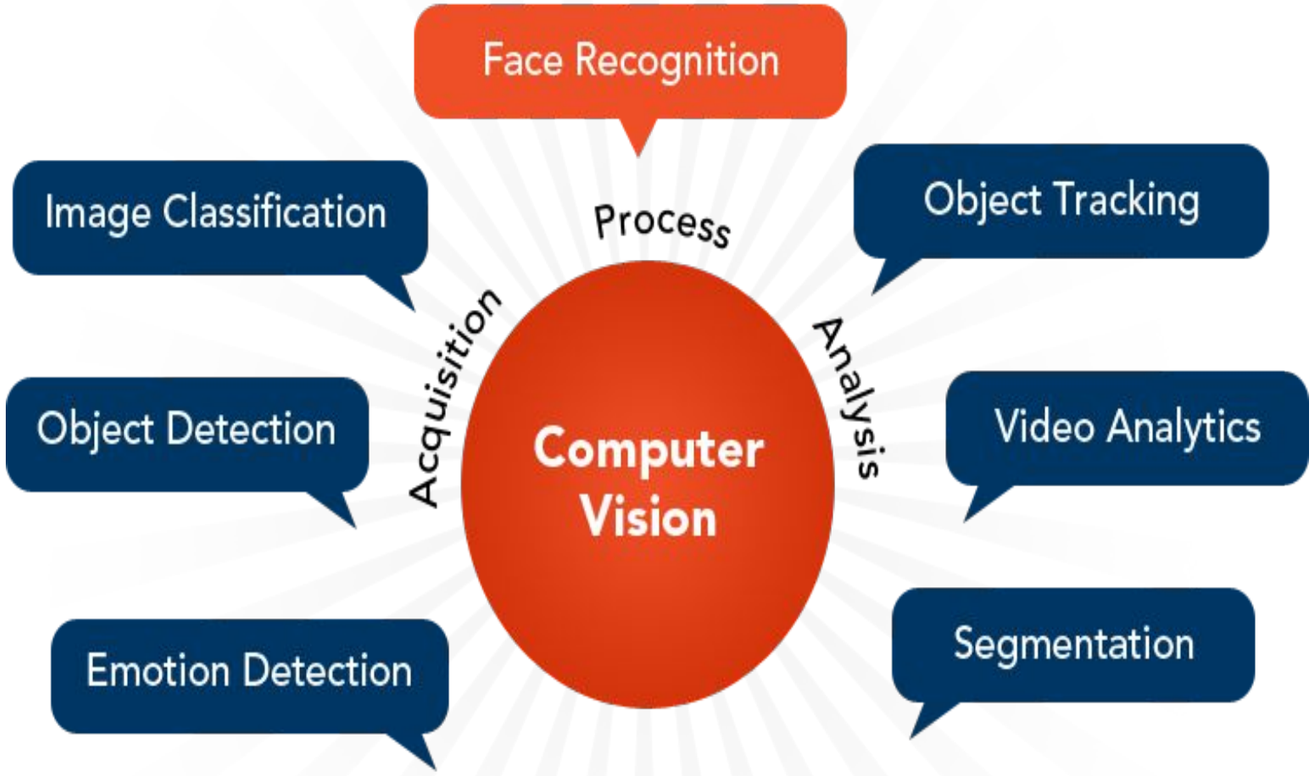


OUTPUT: VEHICLE STOPS

Processing Components

Processing Components are:

1. **Image Acquisition**
2. **Image Processing**
3. **Image Analysis**



A. Image Acquisition

1. Image Classification: Image classification is where a computer can analyze an image and identify the image. Image classification involves **assigning a label** to an entire **image or photograph.**

- Image classification with deep learning most often involves convolutional neural networks, or CNNs.

Some examples of image classification include:

1. Labeling an x-ray as cancer or not (binary classification).

2. Classifying a handwritten digit (multiclass classification).

3. Assigning a name to a photograph of a face (multiclass classification).

IMAGE CLASSIFICATION – EXAMPLES

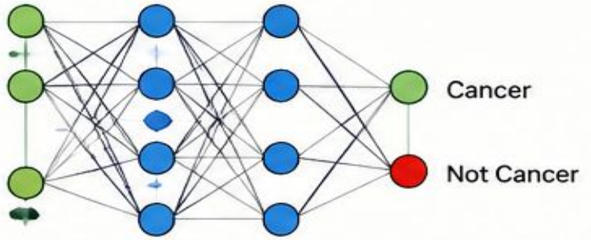
1 Labeling an X-ray as Cancer or Not

(Binary Classification)

Input (X-ray Image)



Deep Learning Model



Output (Predicted Label)



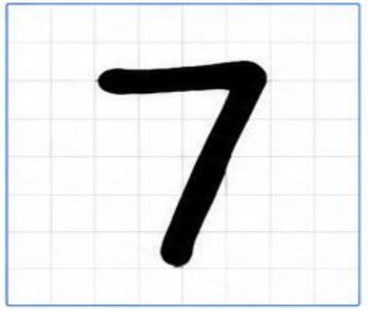
CANCER
or
NOT CANCER

(Binary Classification)

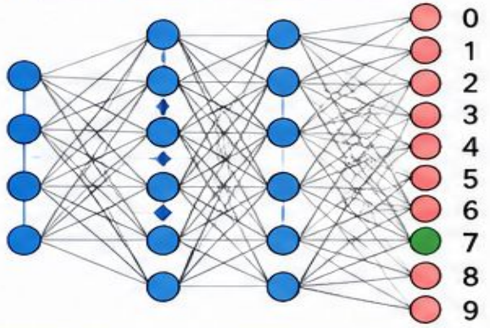
2 Classifying a Handwritten Digit

(Multiclass Classification)

Input (Digit Image)



Deep Learning Model



Output (Predicted Label)



Predicted Digit:
7

(Multiclass Classification)

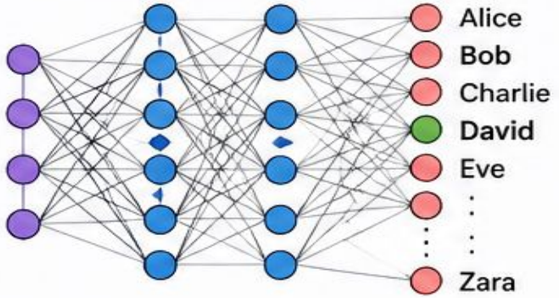
3 Assigning a Name to a Photograph of a Face

(Multiclass Classification)

Input (Face Image)



Deep Learning Model



Output (Predicted Label)



Predicted Name:
David

(Multiclass Classification)

A. Image Acquisition

2. Object Detection: It would be very useful if **robots** can **identify users** by their **faces as humans do**. Face verification system can be used very widely in tasks such as user verification for semi-personal devices, crime investigation, etc.

- **Convolutional Neural Network** is the most promising technique for **extracting features** and **identify the objects of the images.**
- The most important thing in **face verification is dealing with similarity measurement** between the **input images and trained images**.

Examples of Object Detection (Faces)

a) Group Photo



Detect and locate all faces in a group photo.

b) CCTV Surveillance



Detect faces from video frames for security and monitoring.

c) Access Control System



Verify the person's identity before granting access.

Face Verification Process

1. Input Image



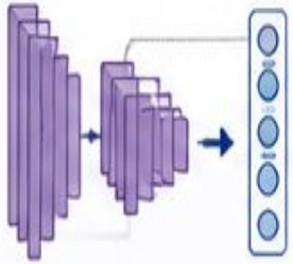
Input face image of the user.

2. Face Detection



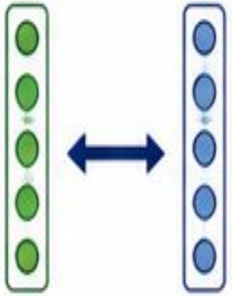
Detect the face in the image.

3. Feature Extraction (using CNN)



Extract deep features from the face.

4. Similarity Measurement



Compare with trained face features.

5. Verification Result

Match

Not Match

Decide whether the faces are same or not.

Important Point

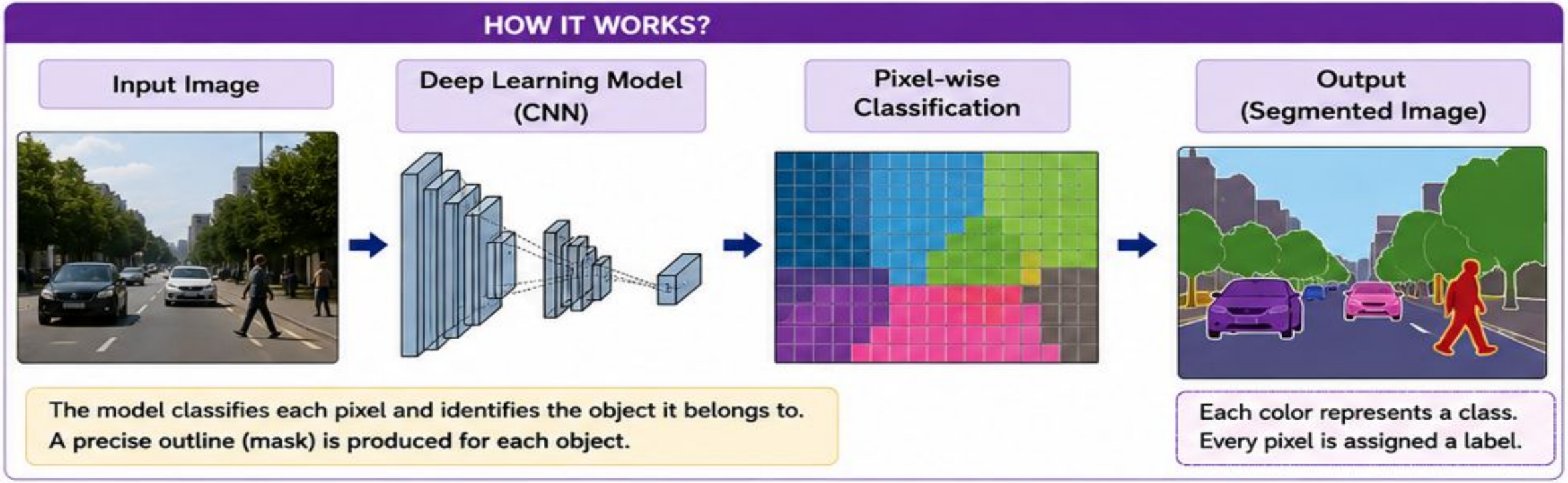
Deep learning (CNN) is the most effective technique for feature extraction and object (face) identification.

Applications

- User verification for devices
- Crime investigation
- Attendance systems
- Access control in organizations
- Smart surveillance systems

3. Object Segmentation: Object segmentation, or semantic segmentation, is the task of object detection **where a line is drawn around each object detected in the image.** Image segmentation is a more general problem of **splitting an image into segments.**

- **Object detection** is also sometimes referred to as **object segmentation.**
- Unlike object detection that involves using a **bounding box to identify objects,** object segmentation identifies the **specific pixels in the image that belong to the object**.



Object Segmentation assigns a class label to every pixel in the image.

1. Self-Driving Car (Street Scene)

Input Image



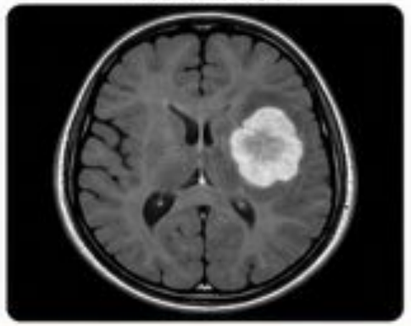
Segmentation Output



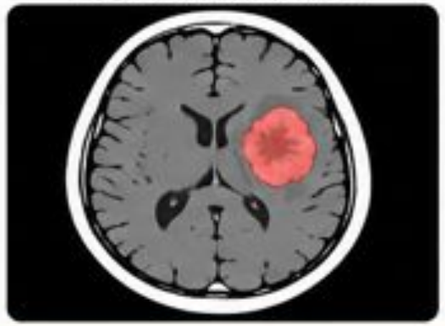
- Road
- Car
- Pedestrian
- Traffic Light
- Tree
- Building
- Sky

2. Medical Image (MRI Brain)

Input Image



Segmentation Output



- Normal Tissue
- Tumor
- Bone
- Background

3. Satellite Image (Land Use)

Input Image



Segmentation Output



- Water
- Forest
- Road
- Buildings
- Bare Land

4. Agricultural Field

Input Image



Segmentation Output



- Healthy Crop
- Diseased Crop
- Soil

5. Human Segmentation (Indoor Scene)

Input Image



Segmentation Output



- Person
- Sofa
- Table
- Lamp
- Wall
- Floor

SEMANTIC SEGMENTATION

Input Image



Output (All cars same class)



INSTANCE SEGMENTATION

Input Image

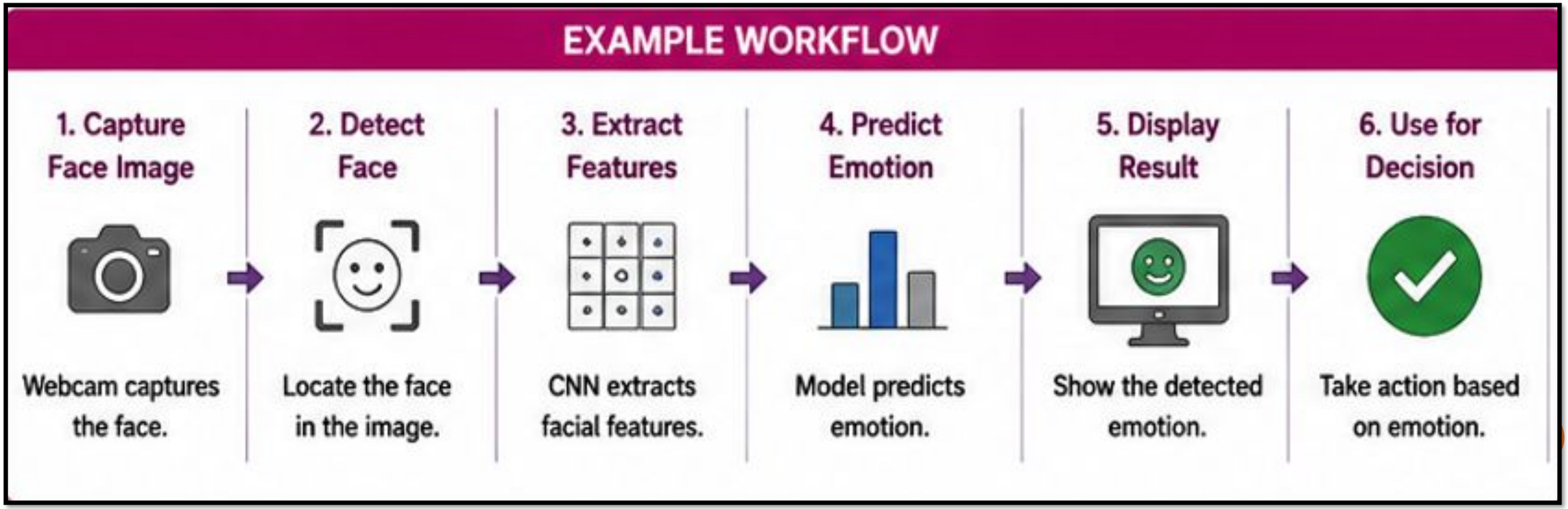


Output (Each car different instance)



4. Emotion Recognition: In this Emotion Recognition we have to used the **purely Convolutional Neural Network to find out Emotion**.

- Here we have to **given the input image of a person**. By using **the CNN we have to find out Emotions of that particular Image.**

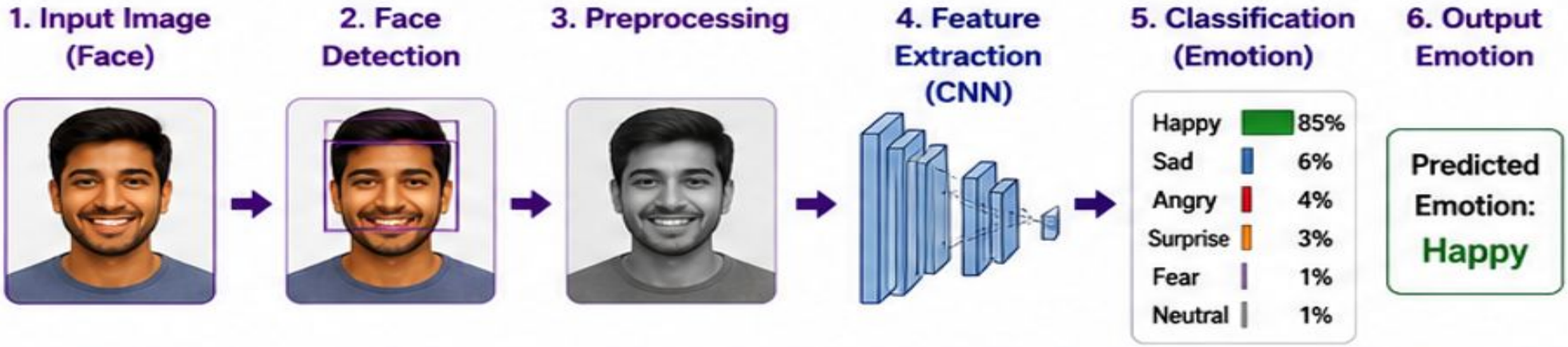


COMMON EMOTIONS DETECTED



The model learns patterns of facial muscle movements and expressions associated with each emotion.

HOW EMOTION RECOGNITION WORKS



5. Image Reconstruction: Image reconstruction and image in **painting is the task** of **filling in missing or corrupt parts of an image.**

- **Examples** include reconstructing **old, damaged black and white photographs and movies (e.g. photo restoration).**
- Datasets often involve using **existing photo datasets** and creating corrupted versions of photos that models must learn to repair.



Image Reconstruction

Image reconstruction (or image inpainting) is the task of restoring missing, damaged, or corrupted parts of an image.

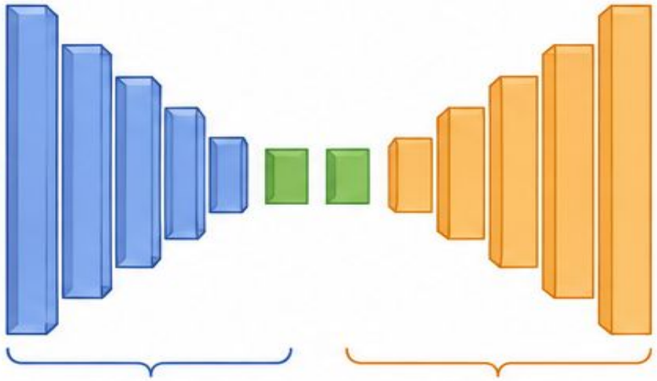
Input Image



Corrupted / Damaged Image



Deep Learning Model (Autoencoder / CNN)



The model learns the context and predicts the missing parts.

Reconstructed Image



Restored / Reconstructed Image

Applications



Old Photo Restoration



Old Movie Restoration



Medical Image Reconstruction



Satellite Image Recovery

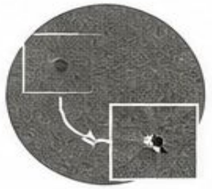


Image Denoising and Inpainting



Art and Painting Restoration

Key Points

- Fills in missing or damaged regions.
- Uses deep learning to understand context.
- Common models: Autoencoders, CNNs, GANs.
- Useful in medical, satellite, movies, and photograph restoration.
- Improves image quality and preserves important information.

B. Process

Facial Recognition: In Facial Recognition we have to find out the **visual Tracking's of the particular Face.**

- Visual object tracking is one of the traditional problems of computer vision and comes to the important core issues with wide-ranging applications **including self-driving car and robot-vision interaction, etc**.
- First, they can consider the **tracking problem** as **classifying each video and frame by learning all dataset**.
- Second, use the deep neural network **as feature generator and use other classifiers** for using their features such as **Artificial Neural Network(ANN).**

STUDENT ATTENDANCE AT CLASSROOM USING FACE RECOGNITION

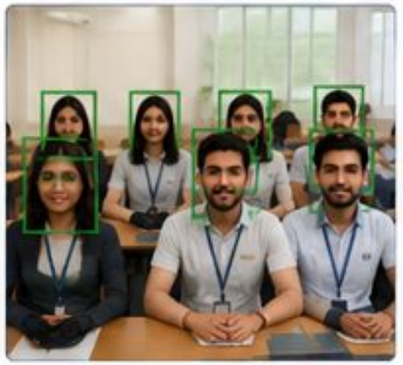
Face Recognition technology is used to automatically mark attendance of students in a classroom.
It is accurate, fast, and prevents proxy attendance.

1. CAPTURE IMAGE



Camera captures live video in the classroom

2. FACE DETECTION



Detects and locates faces in each frame

3. FEATURE EXTRACTION

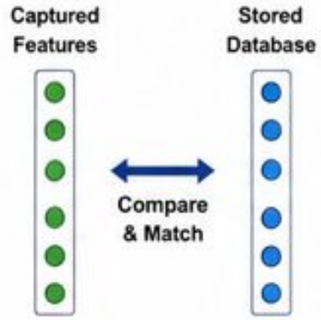


Deep Learning Model (Face Embedding)

[0.12, 0.85, 0.36, ..., 0.71]
[0.35, 0.66, 0.21, ..., 0.54]
...
[0.91, 0.48, 0.75, ..., 0.32]

Extracts unique facial features (embeddings) for each face

4. FACE MATCHING



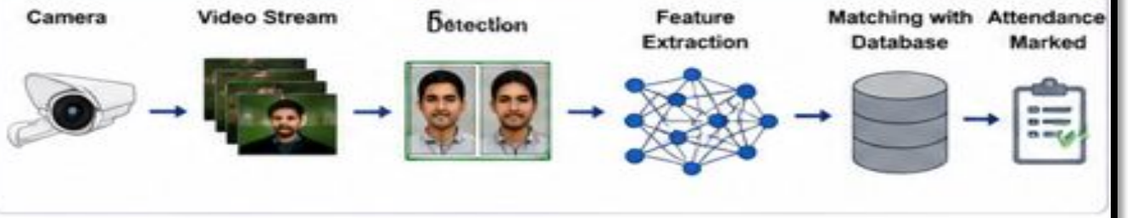
Compares with stored student database

5. MARK ATTENDANCE

Student ID	Name	Time	Status
101	Anjali Sharma	09:15:08	Present
102	Rohit Verma	09:15:09	Present
103	Priya Singh	09:15:10	Present
104	Amit Kumar	09:15:11	Present
...			

Attendance is marked automatically with timestamp

SYSTEM ARCHITECTURE



DATABASE CREATION (ONE TIME)



ATTENDANCE REPORT

Date: 26-05-2025 Class: CSE-3A Time: 09:15 AM



Excel Report PDF Report

C. Analysis

1. Object Tracking: Object tracking is a key step in computer vision for **video surveillance, public safety, and traffic analysis.**

- Object detection and tracking are the two correlated **components of Video Surveillance**.
- Object detection in videos is the first step before performing **complicated tasks such as tracking**.
- **Deep learning neural networks is a powerful programming paradigm** which learns multiple levels **of representation and abstraction of data such as images, sound, and text.**

NUMBER PLATE IDENTIFICATION IN A CAR

Automatic Number Plate Recognition (ANPR) system detects, extracts and reads the number plate from a vehicle image or video.

1. INPUT IMAGE
Capture image/video of the vehicle



2. VEHICLE DETECTION
Detect and locate the vehicle



3. NUMBER PLATE DETECTION
Detect and locate the number plate



4. PLATE EXTRACTION
Extract the number plate region



5. IMAGE PREPROCESSING
Convert to grayscale, resize, threshold



6. CHARACTER SEGMENTATION
Segment characters from the plate



7. CHARACTER RECOGNITION (OCR)
Recognize characters using OCR

DL8CAX1234

8. OUTPUT
Number plate identified

DL8CAX1234

EXAMPLE WORKFLOW

Input Image	Vehicle Detected	Plate Detected	Plate Extracted	Preprocessed Plate	Segmented Characters	Recognized Number
			DL 8C AX 1234	DL 8C AX 1234	DL8CAX1234	DL8CAX1234

APPLICATIONS



Toll Collection



Parking Management



Traffic Monitoring



Vehicle Access Control



Law Enforcement

KEY FEATURES

- Accurate number plate detection and recognition
- Works in real-time on images and videos
- Supports different plate formats
- Robust to varying lighting and angles
- Saves time and reduces manual effort

TECHNOLOGIES USED

- OpenCV (Image Processing)
- YOLOR / YOLOv5 / SSD (Plate Detection)
- Tesseract OCR (Character Recognition)
- Python, NumPy, EasyOCR
- Deep Learning & Computer Vision

SAMPLE OUTPUT

Detected Number Plate: DL8CAX1234 | **Confidence Score:** 98.67% | **Date & Time:** 26-05-2025 10:15:30 AM | **Location:** Vijayawada, AP


Thank You!
Bridging Human Intuition
with Machine Intelligence
Together, We See More.
HUMAN VISION
COMPUTER VISION